



Republic of the Philippines
Department of Agriculture
BUREAU OF FISHERIES AND AQUATIC RESOURCES
REGIONAL FISHERIES LABORATORY XII
J. Catolico St., Lagao, General Santos City

Document Type Work Instruction	Revision No: 0	Date Adopted: 13 Aug 2019	Page No: Page 1 of 6
Document Code: WI-01-02	IN-HOUSE VALIDATION OF HISTAMINE		

Introduction

This document gives the procedures for the in-house validation of histamine.

The parameters to be evaluated are the following:

- Limit of Detection
- Limit of Quantification
- Recovery
- Repeatability
- Reproducibility
- Working range/Linearity

1. Responsibility

- Laboratory Analyst
- Technical Manager - Chemistry

2. Procedure

2.1. Introduction and Purpose

A method validation plan will be created.

It consists of the introduction which includes the following:

- Method
- Duration of the work
- Analysts performing the method validation
- Method scope
- Status of the method
- Analyte
- Measurement unit
- Type of sample and intended use

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- Sampling and subsampling
- Purpose

2.2. Performance Characteristics

A. Validation Parameters

	Description	Procedure
LOD	Test 10 sample blanks, calculate the concentration (ppm), determine the standard deviation and multiply by 3 Reference: EURACHEM Guide Fitness for Purpose 2 nd Edition. Sec 6.2 <i>Limit of Detection and Limit of Quantification</i>	• Add 40mL of methanol into 50mL polypropylene tube (pp). • Perform analysis • Calculate the concentration of each trial from the curve. • Calculate the standard deviation and multiply by 3. • This will be done in 10 trials on separate occasions. LOD: 3 x SD
LOQ	Test 10 sample blanks, calculate the concentration (ppm), determine the standard deviation and multiply by 3 Reference: EURACHEM Guide Fitness for Purpose 2 nd Edition. Sec 6.2 <i>Limit of Detection and Limit of Quantification</i>	• Add 40mL of methanol into 50mL polypropylene tube (pp). • Perform analysis • Calculate the concentration of each trial from the curve. • Calculate the standard deviation and multiply by 3. • This will be done in 10 trials on separate occasions. LOQ: 10 x SD
Recovery	• Test the recovery in spiked samples • Calculate the % Recovery	• Weigh 5.0 g of sample into 50mL PP tube • This will be done in 4 trials, 2 trials for spiking and the other 2 trials for sample blank.

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	Target value: <u>80-110%</u> (refer to AOAC Appendix F, p.9, Table A5)	- For spiked duplicate samples add 50µL each of 1000ppm histamine standard (expected recovery: 10 ppm) - Analyze the samples - Perform in at least 7 different occasions (or 7 trials) - Calculate ppm concentration of each sample present from the curve. - Calculate the average concentration of blank samples and spiked samples. - Using the average concentration, calculate the % Recovery Recovery = (Recovered Histamine / Histamine in Spiked Sample) x 100
Repeatability	- Test at least 10 sets of samples of identical matrix. - Analysis will be performed by the same analyst. Target value: CV = 15% for analyte conc. <1 ppm CV = 10% for analyte conce. >1 ppm (Refer to EU Commission decision 2002/657/EC, Table 8)	- Weigh 5.0 g of sample into 50mL pp tube. - Analyze the samples on the same day. - Calculate the ppm concentration of each sample present from the curve. - Calculate the mean, standard deviation and the (%) coefficient of variation using the concentration present in each samples.
Reproducibility	- Test at least 7 sets of samples of identical matrices. - Analysis will be performed by different analyst in the same day	- Weigh 5.0 g of sample into 50mL pp tube. - This will be done in 4 trials, 2 trials for spiking and the other 2 trials for sample blank.

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	Target value: <u>80-110%</u> (refer to AOAC Appendix F, p.9, Table A5) Target value: CV = <u>7.3%</u> (refer to AOAC Appendix F, p.9, Table A4)	- For spiked duplicate samples add 50µL each of 1000ppm histamine standard (expected recovery: 10ppm) - Analyze the samples - Perform in at least 7 different occasions (or 7 trials) - Calculate the ppm concentration of each sample present from the curve. - Calculate the average concentration of blank samples and spiked samples. - Using the average concentration, calculate the % Recovery. - Compute the average mean, standard deviation and %CV between the two analysts.
Working range/Linearity	Prepare 6 calibration points: 0 ppm, 0.01 ppm, 0.05 ppm, 0.1 ppm, 0.2 ppm and 0.3 ppm Target value: $R^2 \geq 0.995$	In every conduct of analysis prepare histamine standard solutions: 0.01 ppm, 0.05 ppm, 0.1 ppm, 0.2 ppm and 0.3 ppm Diluent: 0.1 N HCl = 0 ppm 0.3 ppm = 3 mL of 10 ppm dilute to 100 mL 0.2 ppm = 2 mL of 10 ppm dilute to 100 mL 0.1 ppm = 1 mL of 10 ppm dilute to 100 mL 0.05 ppm = 500 µL of 10 ppm dilute to 100 mL 0.01 ppm = 100 µL of 10 ppm dilute to 100 mL Linearity is justified if the R^2 value is greater than or equal to 0.995.

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B. Time Table:

Validation Parameters	Day									
	1	2	3	4	5	6	7	8	9	10
LOD	X	X	X	X	X	X	X	X	X	X
LOQ	X	X	X	X	X	X	X	X	X	X
Recovery	X	X	X	X	X	X	X			
Repeatability	X									
Reproducibility	X	X	X	X	X	X	X			
Working range/Linearity	X	X	X	X	X	X	X	X	X	X

3. Outline of the Experiment

3.1. Sample Preparation

Refer to WI-01-CHE-01, Sample Preparation (Chemistry).

3.2. Preparation of Reagent Blank and Spiked Samples

Refer to WI-01-CHE-02, Histamine (section 7.2).

3.3. Experiment Procedure and Calculation

Refer to WI-01-CHE-02, Histamine (section 7.3 -8.0).

4. Documentation and Reporting

4.1. Recording of Results

Data from all validation experiments will be recorded in the computer used for histamine analysis, with Microsoft Excel as the interface for the Trilogy software.

4.2. Summary Report

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A validation report will be created, which summarizes the validation work and results. This includes raw data, statistical analysis and conclusion with an evaluation of whether the method is fit for purpose.

5. Revalidation

The method will be revalidated periodically or when changes to personnel, equipment, reagents, or sample matrices occur.

6. Document History

Revision No.	Issue Date	Original Entry	Revised as	Recall Date

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